Progress Report

Thierry Sae-Lim – Norikazu Takahashi
Okayama University
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Research Objective and Goal
The final goal of the project is to implement the distributed algorithm for PCA (Principal Component Analysis) developped by our lab and test its validity based on numerical experiment.

Short-Term Goals
10. Continue studying the number of iteration *T_Y* and *T_Z* in the averaging consensus of the matrices *Y_m(t)* and *Z_m(t)* and the number of iteration *T_PM* in the distributed algorithm for PCA alongside with star graphs, path graphs, circle graphs, and regular graphs.

Progress

We recall that we have shown differents convergence table for a convergence stopping condition equals to 1e-8, 1e-9 and 1e-10 depending on T_Y and T_Z.

We will now plot the minimal number of iterations T_Y and T_Z depending on the algebraic connectivity of the agent network. To this end, we will generate other graphs and find the bests associated number of iteration T_Y and T_Z, that is to say, the numbers minimizing the overall number of iterations.

To save time, we will create a searching algorithm to find the bests parameters T_Y and T_Z. The algorithm is based on a gradient descent algorithm.

To simplify the algorithm's explanation, let's say that we are on a convergence table grid. We start from predefined numbers y and z and compute the overall number of iterations needed with T_Y=y and T_Z=z and its 8 directional neighbors. We next save the results to avoid recomputing what has already been computed. Then we take the numbers y and z associated with the minimal number of iterations needed computed. But this would always require to compute a lot of times the decentralized algorithm for PCA as we will compute a bunch of neighbors as we update the parameters y and z. To reduce it and save time, if we find another neighbor that needs a less overall number of iterations, we immediately update the parameters y and z and start the loop over. However, the bests values of T_Y and T_Z can be far from the initial y and z parameters. We can save more time by variating the update step for y and z.

To give and example, we can compute for the 8 directional neighbors of y and z for a step of x0. This means that we will computes for the following coordinates in the convergence table grid:
(y+x0, z), (y, z+x0), (y+x0, z+x0), (y-x0, z), (y, z-x0), (y-x0, z-x0), (y+x0, z-x0), (y-x0, z+x0)
Then when we find the optimum values for a step equals to x0, we can reduce the step number to x1 < x0 and start over until xn = 1 to find the optimum values of T_Y and T_Z.

Now that we can find the bests values of T_Y and T_Z a little bit quicker, we will generate graphs with 8 agents and variating the number of edges (7, 10, 13, 16, 19, 22, 25, 27) :

[see printed documents]

However, a mistake was made during the generations of the instances: They were all independantly generated. Thus, the showed results were influenced not only by the agent network structures but also by the randomness of the instances. To solve this problem, we will now only generate a single data matrix X, and set predefined initial vectors defined as:

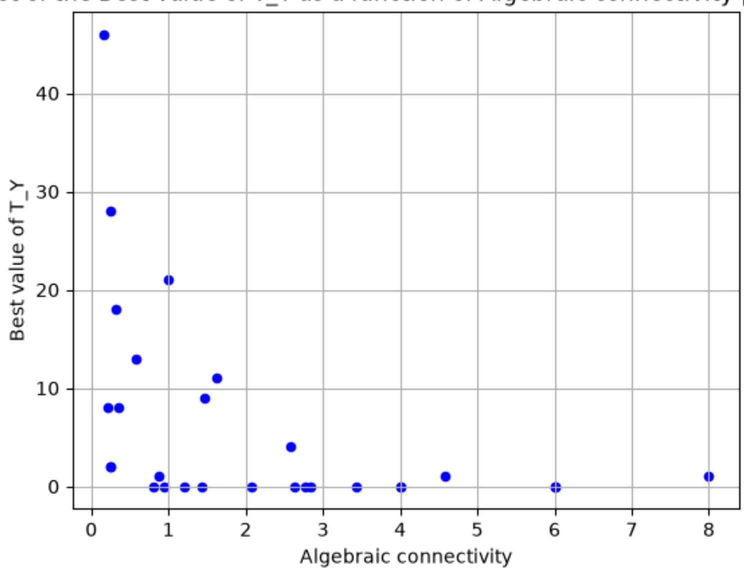
u_p(0) = e_p , with p=1,…,P and e_p as the p-th vector of the canonical basis.

We can now restart the process with this more accurate experiment:

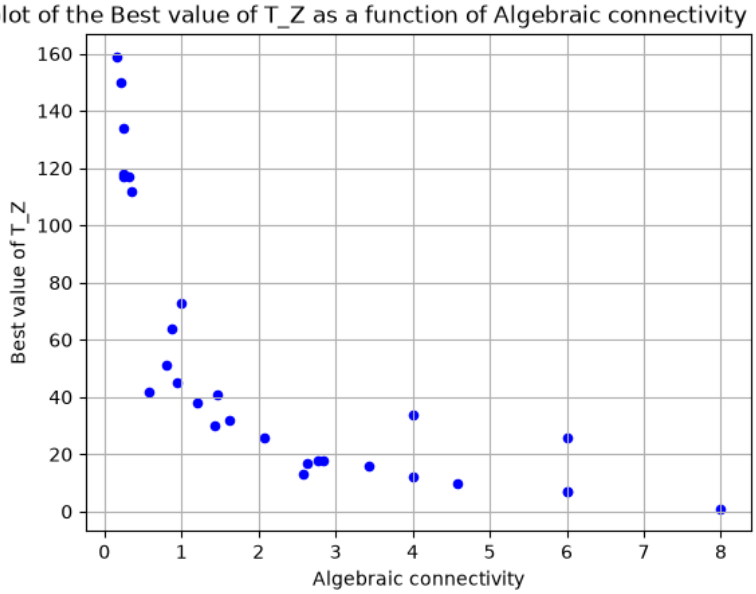
[see printed documents]

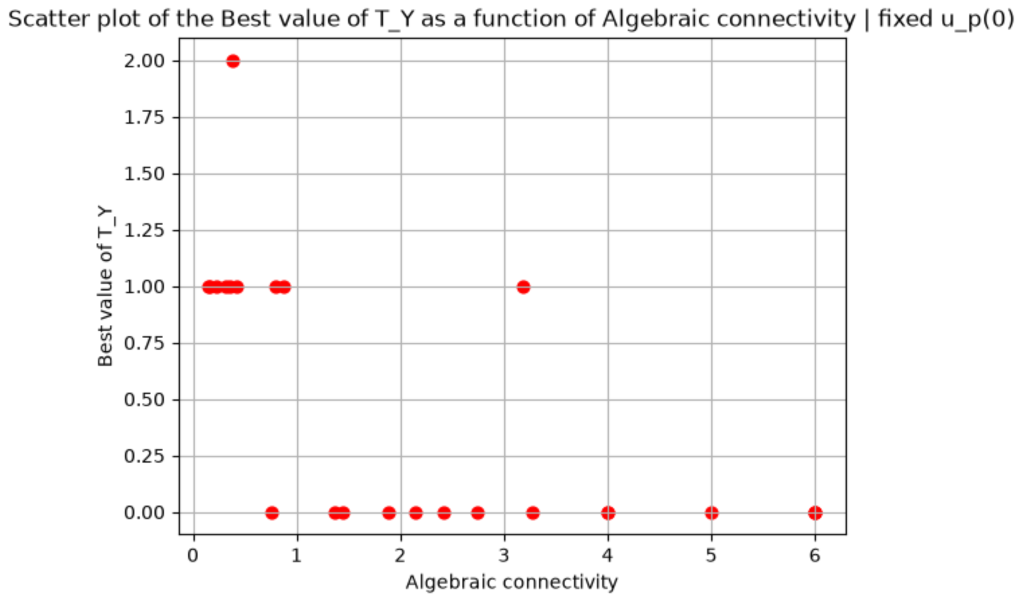
Now that we have several point, we can now display scatter plots:

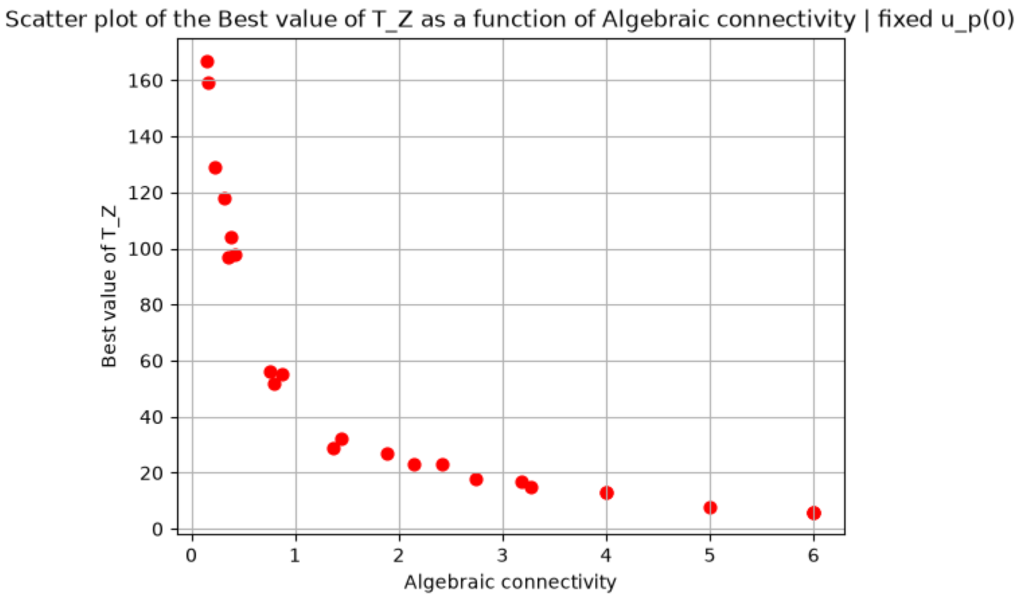
Scatter plot of the Best value of T_Y as a function of Algebraic connectivity | random u_p(0)

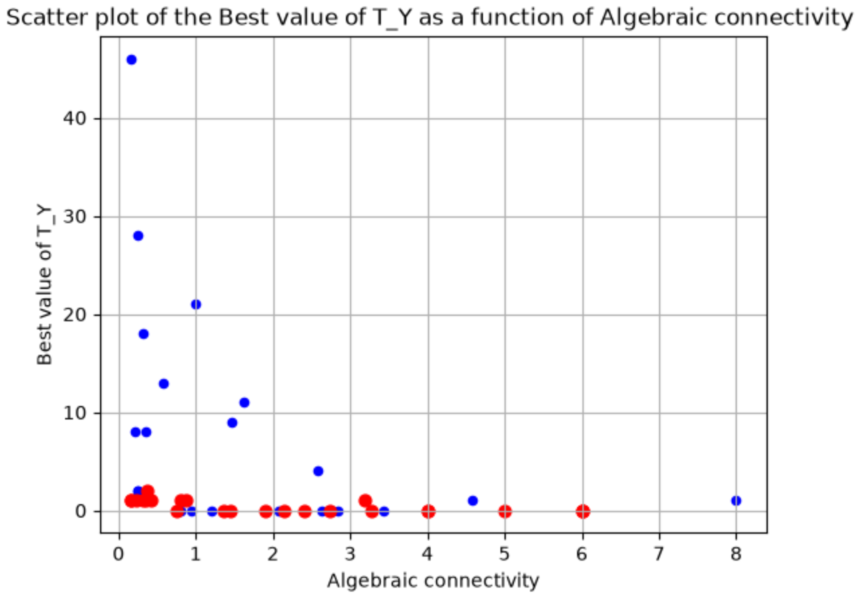


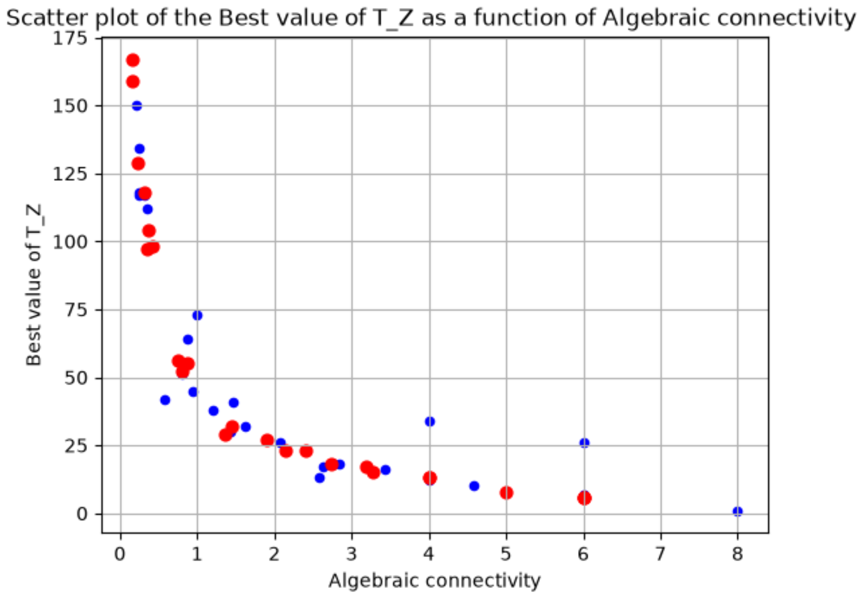
Scatter plot of the Best value of T_Z as a function of Algebraic connectivity | random u_p(0)




Scatter plot of the Best value of T_Y as a function of Algebraic connectivity | fixed u_p(0)
Best value of T_Y
Algebraic connectivity
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
1
2
3
4
5
6


Scatter plot of the Best value of T_Z as a function of Algebraic connectivity | fixed u_p(0)
Best value of T_Z
Algebraic connectivity
160
140
120
100
80
60
40
20
0
0
1
2
3
4
5
6





We can clearly see a relation beteween the best value of T_Z and the algebraic connectivity.
The relation between the best value of T_Y and the algebraic connectivity is less obvious. It may be also influenced by other parameters such as the spectral radius of the weight matrix ρ(W).